Reference sheet

Vectors

Vector components	Resultant vectors
$A_x = A\cos\theta$ $A_y = A\sin\theta$ θ is measured from the +x axis. Left and down components are negative.	Add components: $R_x = A_x + B_x, R_y = A_y + B_y$ Magnitude: $R = \sqrt{R_x^2 + R_y^2}$ Direction: $\theta = \tan^{-1}\left(\frac{\lvert R_y\rvert}{\lvert R_x\rvert}\right)$

SOH-CAH-TOA: $\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}}, \cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}, \tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$

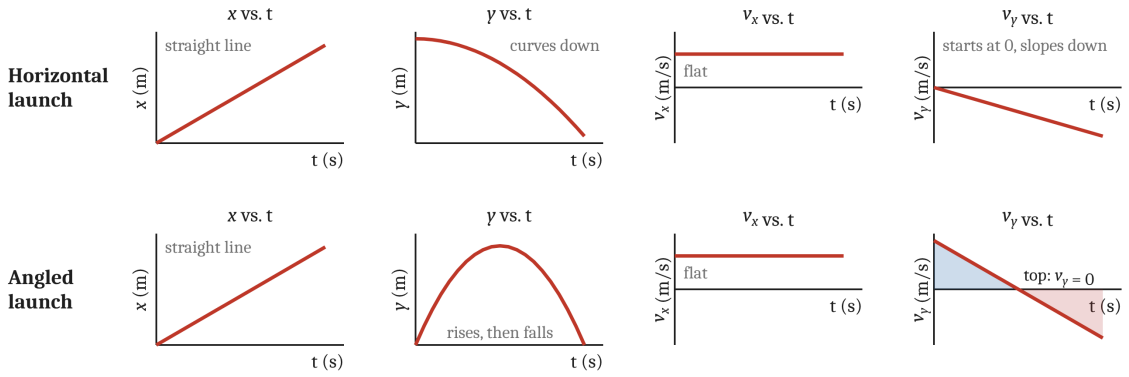
Pythagorean theorem: $c^2 = a^2 + b^2$

Relative velocity

- Add velocities as vectors: $v_{\text{object rel. ground}} = v_{\text{object rel. medium}} + v_{\text{medium rel. ground}}$
- **Assume:** each velocity is constant, and the medium (water or air) carries the object along with it.
- **Perpendicular velocities:** find the resultant with the Pythagorean theorem and its direction with $\tan^{-1}$.
- **Crossing time:** only the velocity component straight across the river sets it.
- **Drift:** the current carries the boat downstream while it crosses.
- **Extension, landing directly across:** aim upstream so the upstream part of the boat's velocity cancels the current.

Motion graphs

Top row: a horizontal launch. Bottom row: an angled launch on level ground. In the angled v_y graph, the shaded area above the axis is the rise and the area below is the fall.



Projectile motion: x world and y world

Ignore air resistance. Down is negative. Time t is the same in both columns and links them.

x world (horizontal)	y world (vertical)
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x = 0$	$a_y = -9.8\ \text{m/s}^2$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
t =	t =
Every projectile	
Assume: $a_x = 0$, so $v_{fx} = v_{ix}$ the whole flight. **Equation:** $x_f = v_{ix}t + x_i$	**Assume:** $a_y = -9.8\ \text{m/s}^2$. At the top, $v_y = 0$. **Equations:** $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$ $v_{fy} = a_y t + v_{iy}$ $v_{fy}^2 = v_{iy}^2 + 2a_y(y_f - y_i)$
Horizontal launch	
Assume: $v_{ix} =$ launch speed, $x_i = 0$.	**Assume:** $v_{iy} = 0$, $y_i = h$ (launch height), $y_f = 0$ (the ground).
Angled launch from the ground	
Assume: $v_{ix} = v_i \cos\theta$, $x_i = 0$. Range uses the total time.	**Assume:** $v_{iy} = v_i \sin\theta$, $y_i = 0$, $y_f = 0$. It lands with $v_{fy} = -v_{iy}$.
Extension: angled launch from a cliff	
Assume: $v_{ix} = v_i \cos\theta$, $x_i = 0$.	**Assume:** $v_{iy} = v_i \sin\theta$, $y_i = h$ (cliff height), $y_f = 0$ (the ground). v_{fy} is negative at landing.

Solving routine

1. Sketch the path. Label the launch point, the landing point, and the ground ($y = 0$).
2. Fill in both columns. Mark what you know and what you want.
3. Find t from the column that has enough knowns (usually the y world).
4. Use that t in the other column to finish.

Common mistakes

- $v_{iy} = 0$ only for a horizontal launch.
- a_x is always 0. Only the y world speeds up or slows down.
- Cosine goes with the side next to the angle: $v_{ix} = v_i \cos\theta$ when θ is measured from the horizontal.
- At the top, $v_y = 0$ but v_x is not, so the ball is still moving.
- v_f is not 0 at the ground. The equations describe the ball just before it lands, while it is still falling fast; the stop happens during the impact, which the equations do not model. Use $y_f = 0$, not $v_{fy} = 0$.

Velocity at any moment

$v_x = v_{ix}$ the whole flight and $v_y = a_y t + v_{iy}$, so $v = \sqrt{v_x^2 + v_y^2}$ and $\theta = \tan^{-1}\left(\dfrac{|v_y|}{v_x}\right)$ (below the horizontal when v_y is negative, above it when positive).